In [210…

```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Make plots appear inside notebook
%matplotlib inline

# Set style for better visuals
sns.set_style("whitegrid")
sns.set_context("talk") # makes text bigger
```

In [211…

```
data = pd.read_csv('https://raw.githubusercontent.com/dphi-official/Datasets/master/t
data
```

Out[211…

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8
...	...	...	...	...	...	...	...	...	...	
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7

891 rows × 12 columns

In [212…

```
titanic.shape          # number of rows and columns
```

```
titanic.info()          # types and missing values
titanic.describe()      # statistical summary
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

Out[212…

	PassengerId	Survived	Pclass	Age	SibSp	Parch	
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.00
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.20
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.69
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.00
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.91
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.45
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.00
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.32

In [213…

```
data.describe(include = 'object')
```

Out[213…

	Name	Sex	Ticket	Cabin	Embarked
count	891	891	891	204	889
unique	891	2	681	147	3
top	Braund, Mr. Owen Harris	male	347082	B96 B98	S
freq	1	577	7	4	644

In [214…

```
data.isnull().sum()
```

Out[214…

```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

In [215…

```
data['Age'] = data['Age'].fillna(np.mean(data['Age']))
```

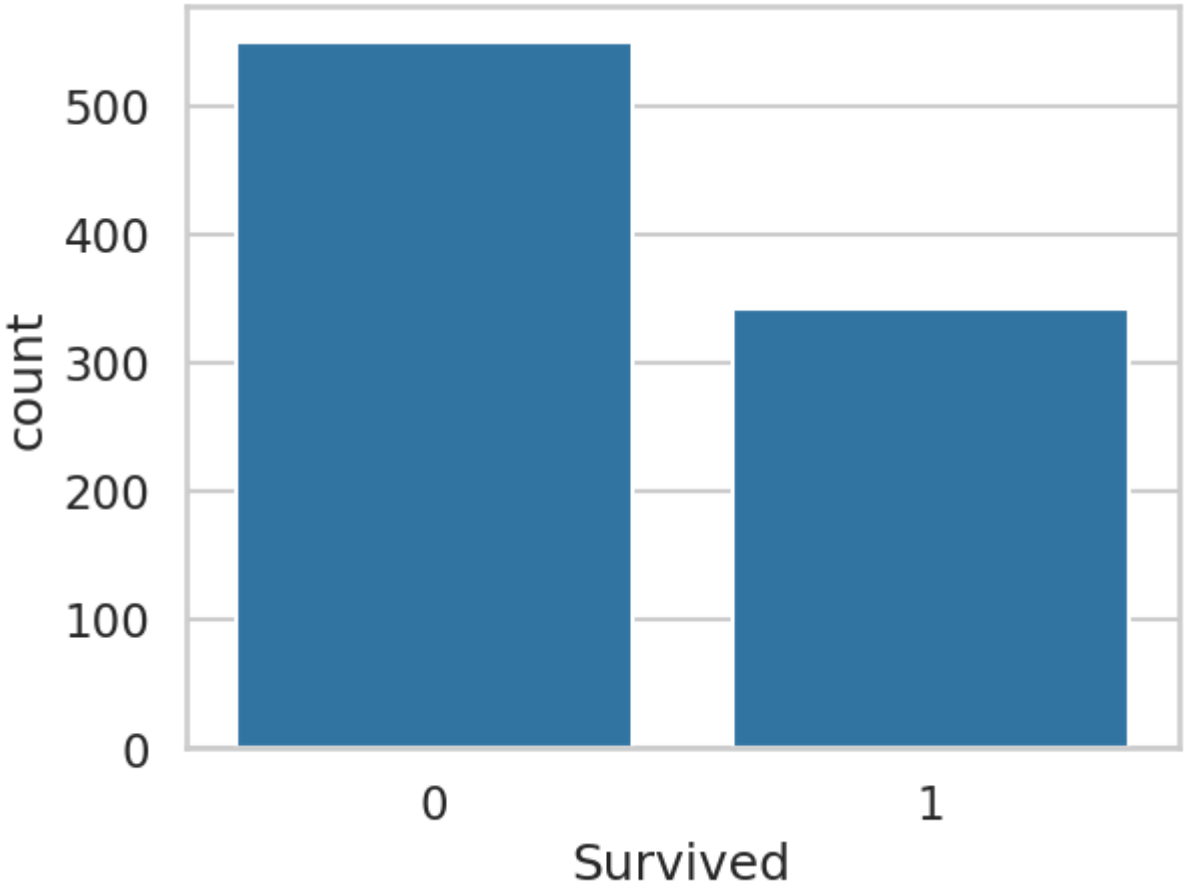
In [216…
```
data['Cabin'] = data['Cabin'].fillna(data['Cabin'].mode()[0])
```

In [217…
```
data['Embarked'] = data['Embarked'].fillna(data['Embarked'].mode()[0])
```

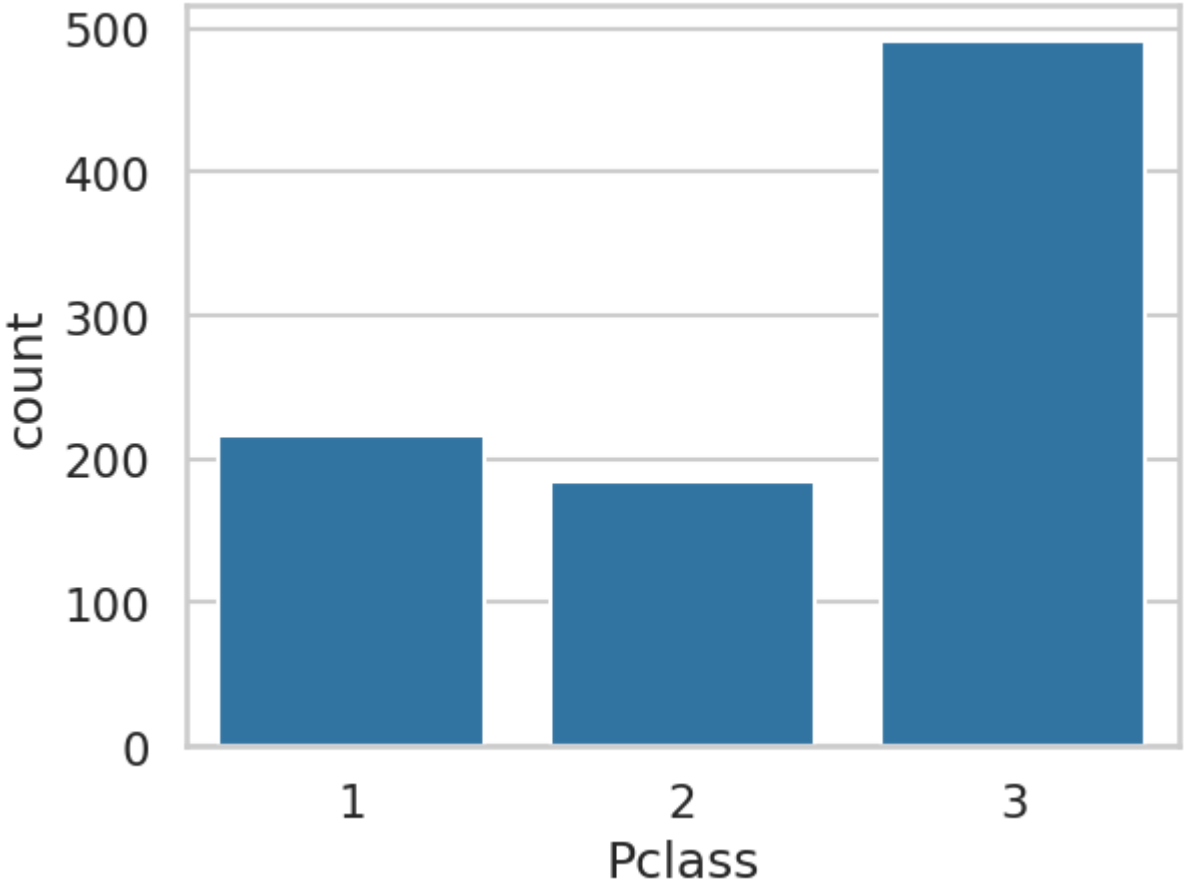
In [218…
```
data.isnull().sum()
```

Out[218…
```
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age            0
SibSp          0
Parch          0
Ticket         0
Fare           0
Cabin          0
Embarked       0
dtype: int64
```

In [219…
```
sns.countplot(x='Survived', data=data)
plt.show()
```

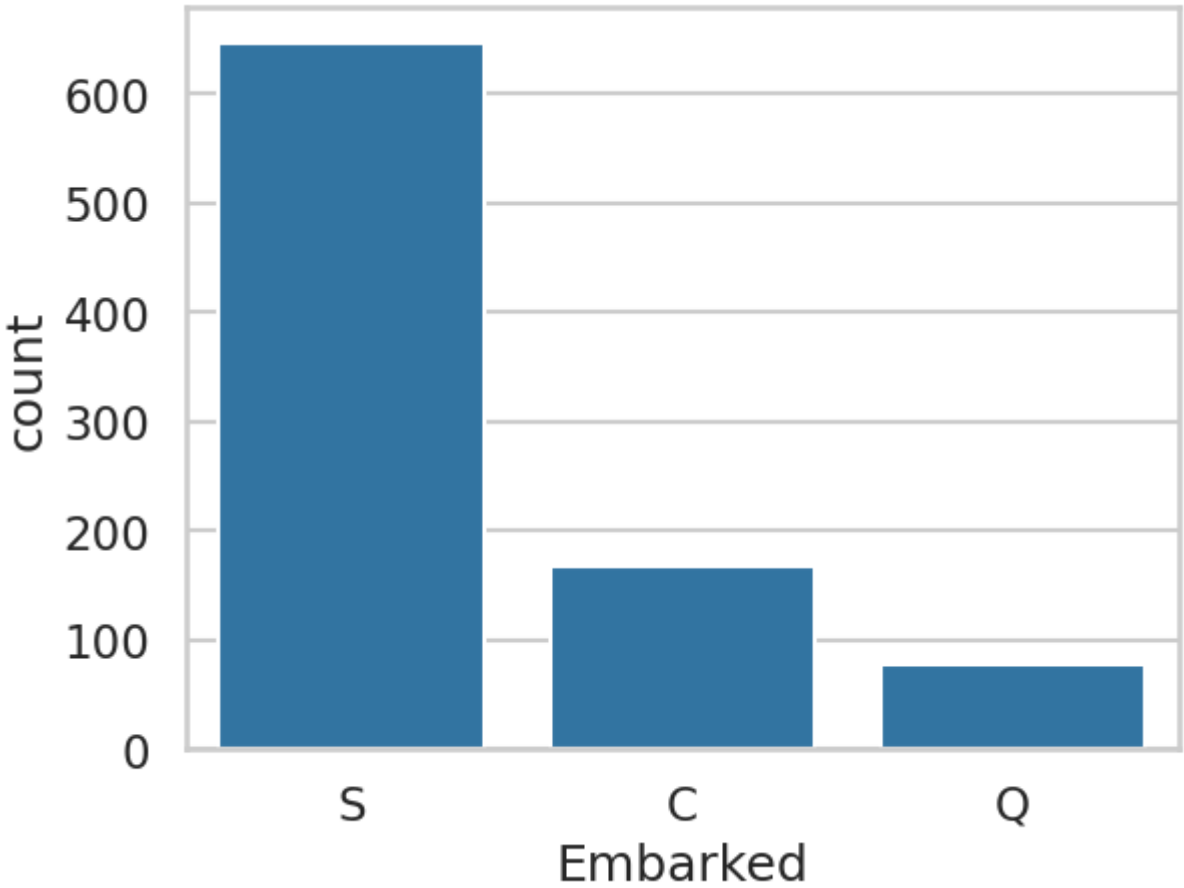


In [220…
```
sns.countplot(x='Pclass', data=data)
plt.show()
```



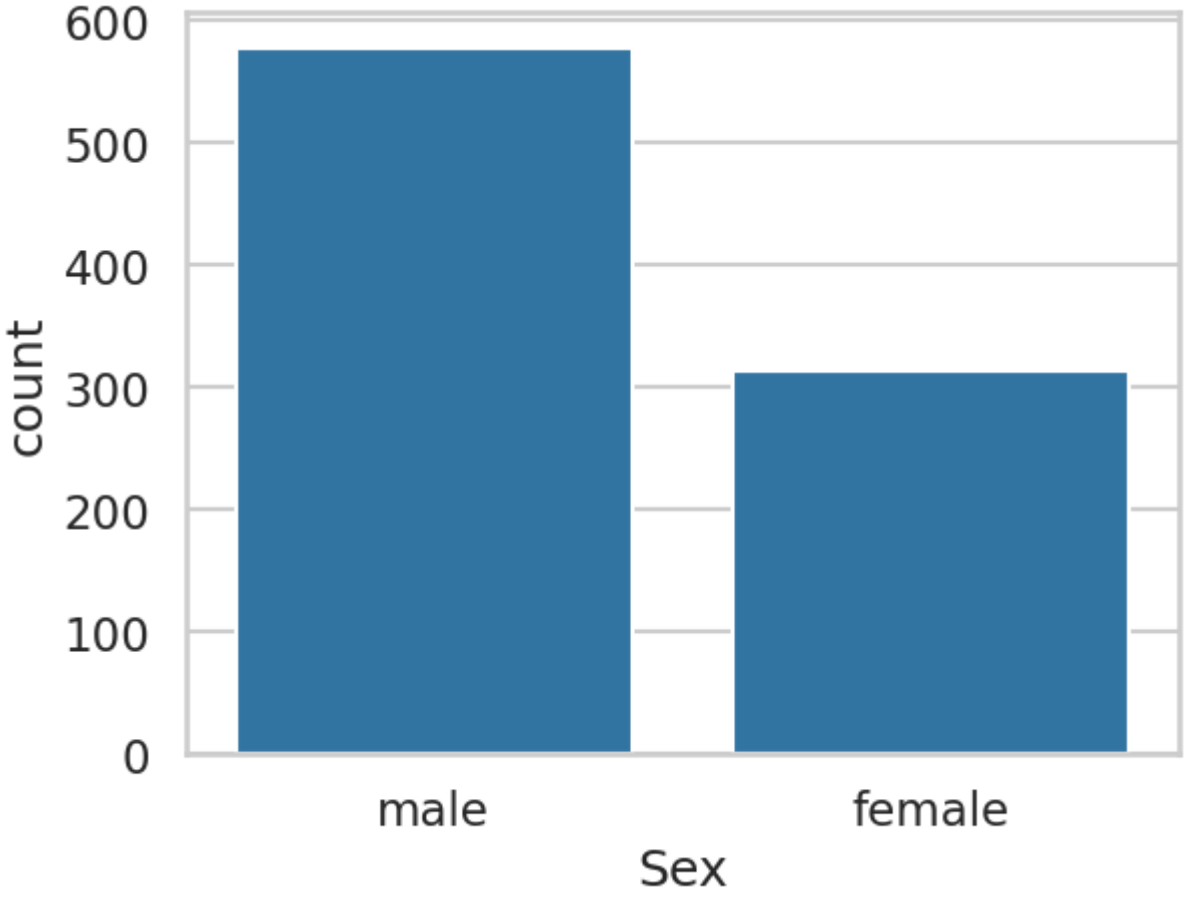
In [221…

```
sns.countplot(x='Embarked' , data=data)
plt.show()
```



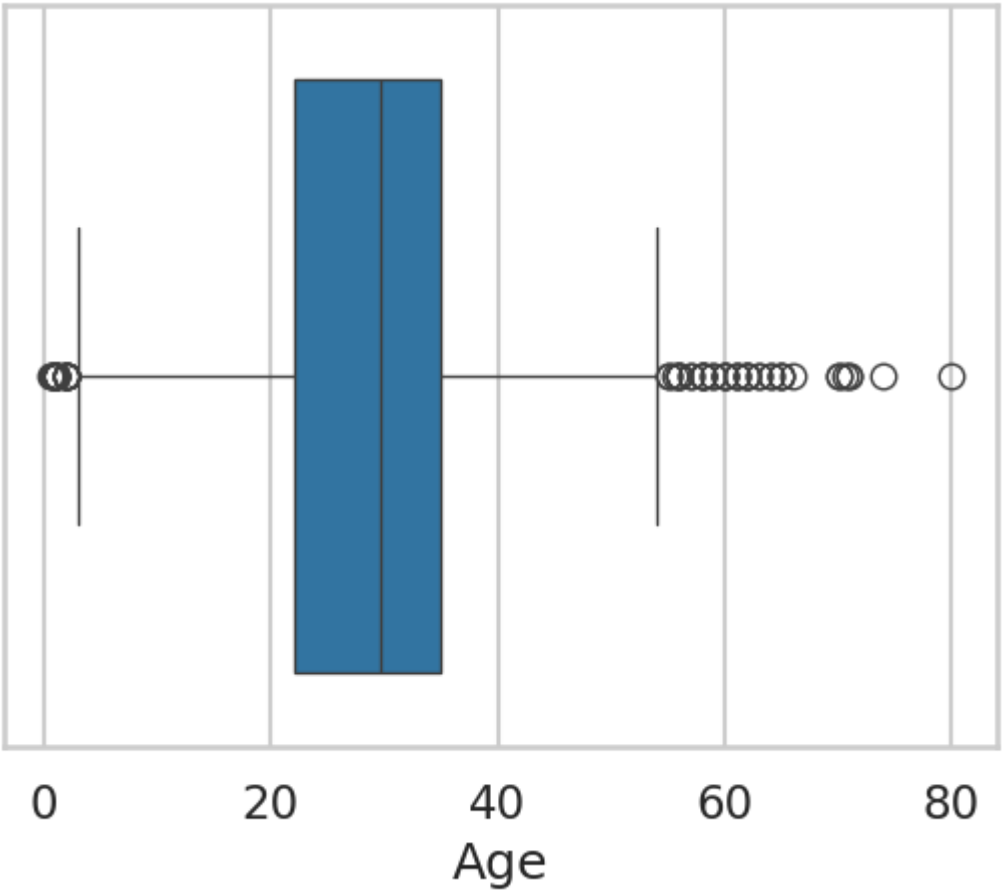
In [222…

```
sns.countplot(x='Sex' , data=data)
plt.show()
```



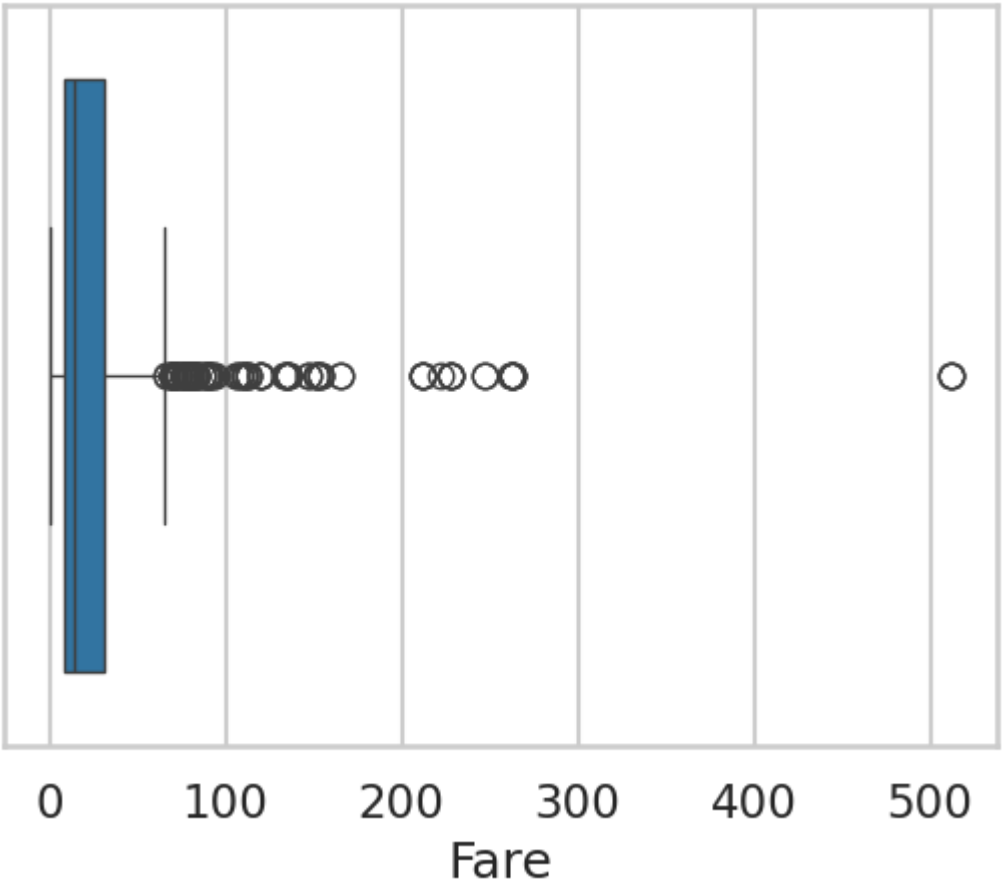
In [223…

```
sns.boxplot(x='Age' , data=data)
plt.show()
```



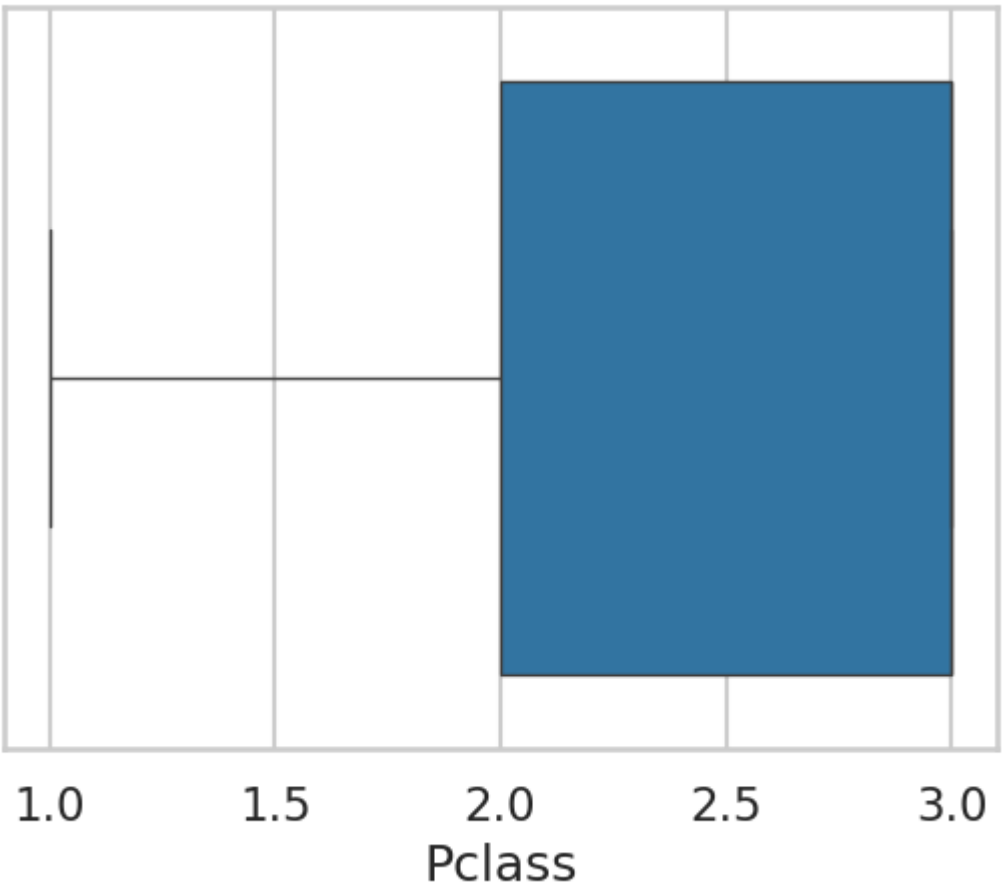
In [224…

```
sns.boxplot(x='Fare' , data=data)
plt.show()
```



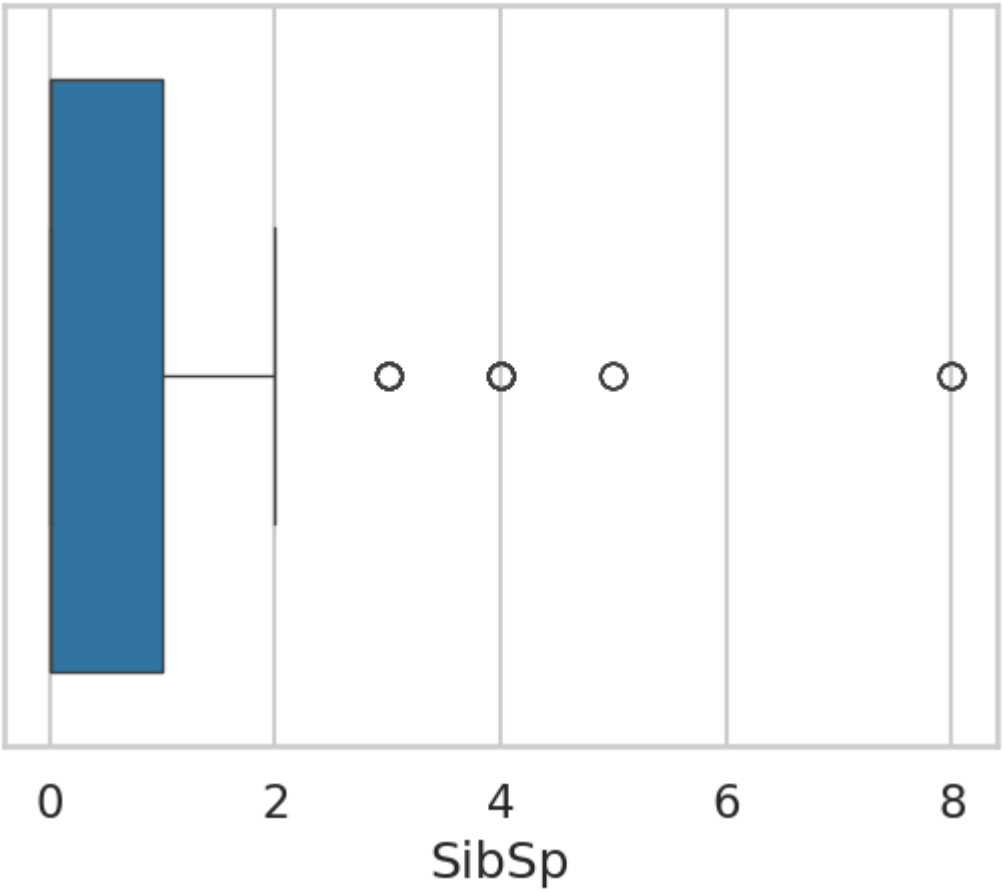
In [225…

```
sns.boxplot(x='Pclass' , data=data)
plt.show()
```



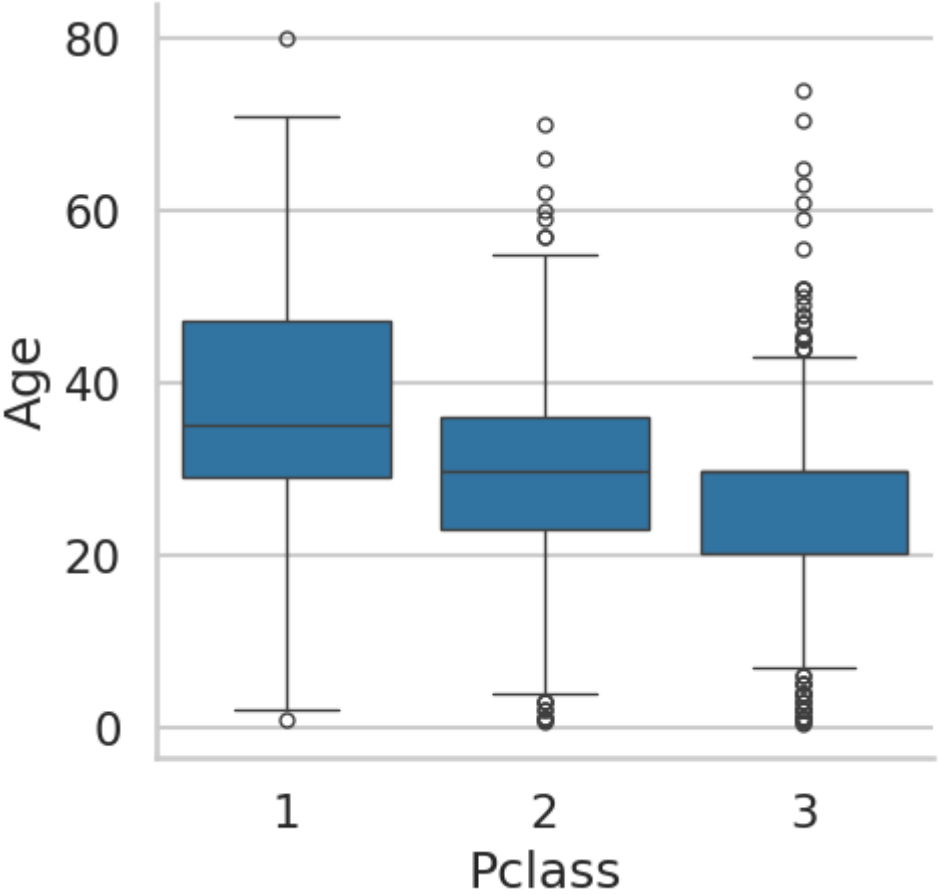
In [226…

```
sns.boxplot(x='SibSp' , data=data)
plt.show()
```



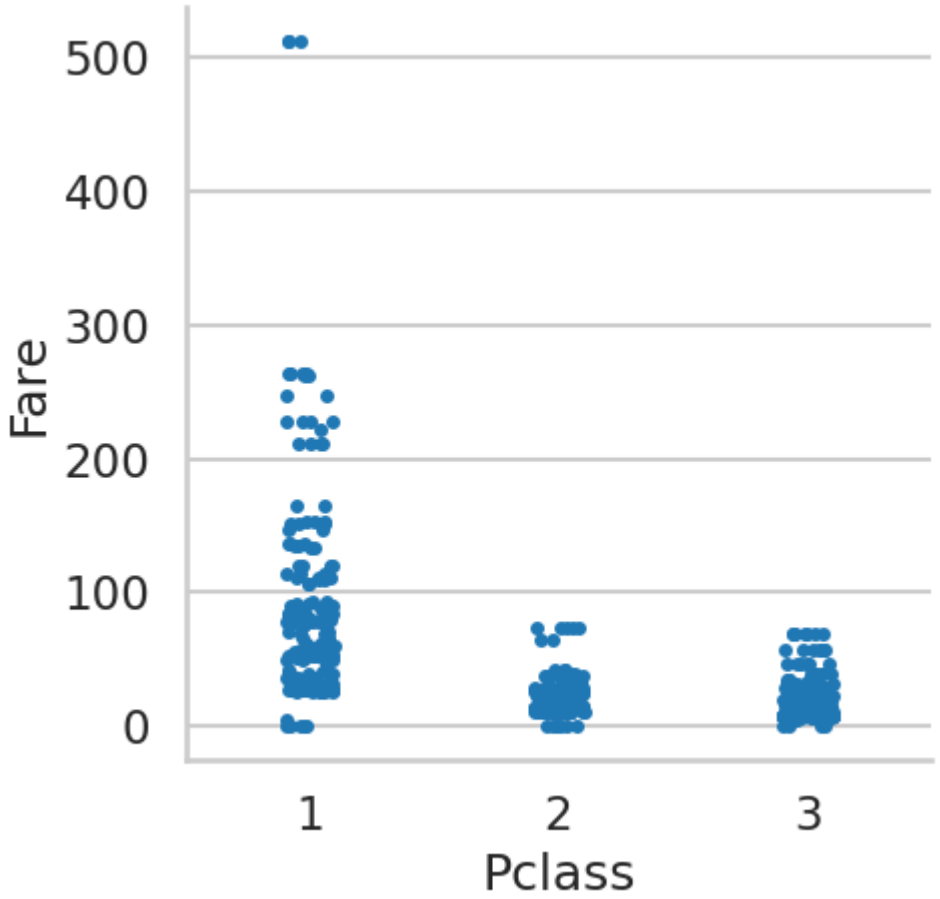
In [227…

```
g = sns.catplot(x='Pclass', y='Age', data=data, kind='box')

plt.show()
```



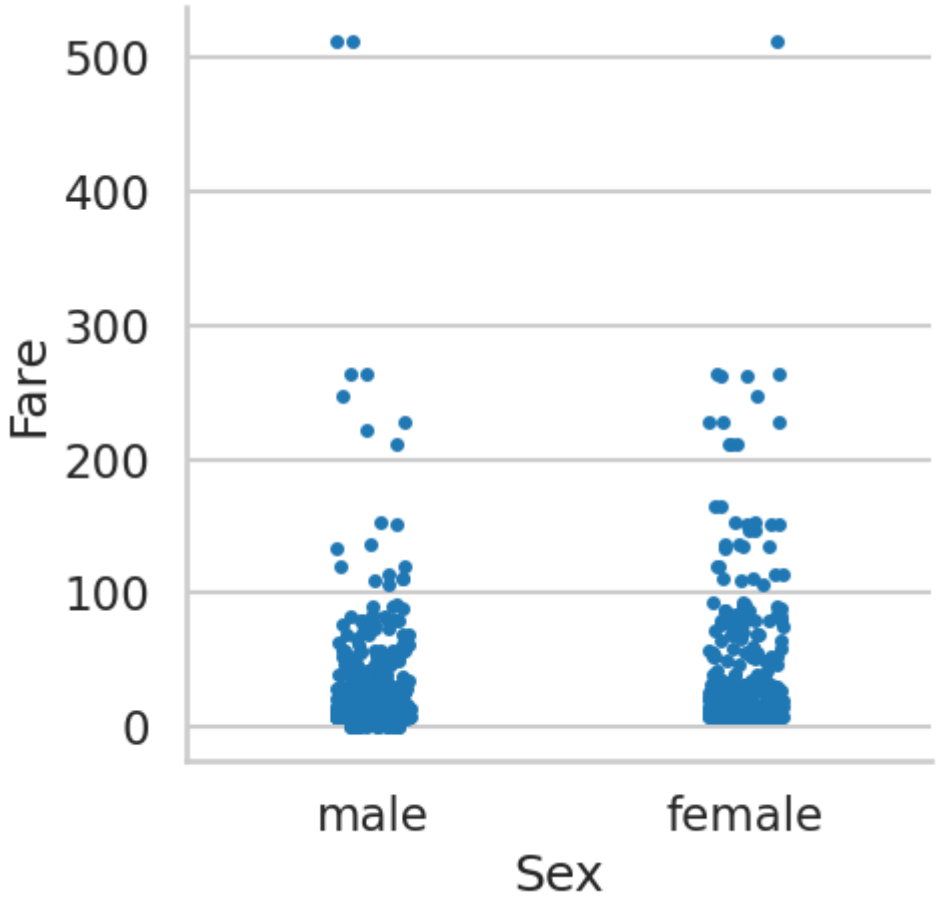
In [228…

```
g = sns.catplot(x= 'Pclass', y = 'Fare', data=data, kind = 'strip')
plt.show()
```



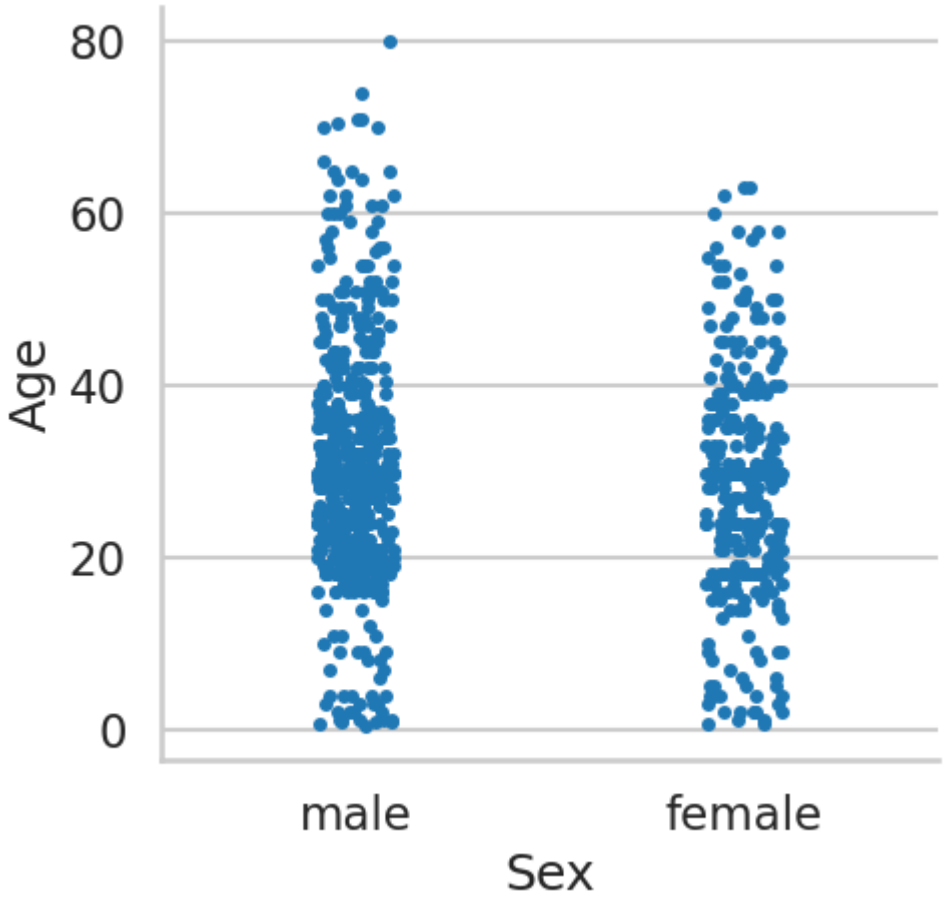
In [229…

```
g = sns.catplot(x= 'Sex', y = 'Fare', data=data, kind = 'strip')
plt.show()
```



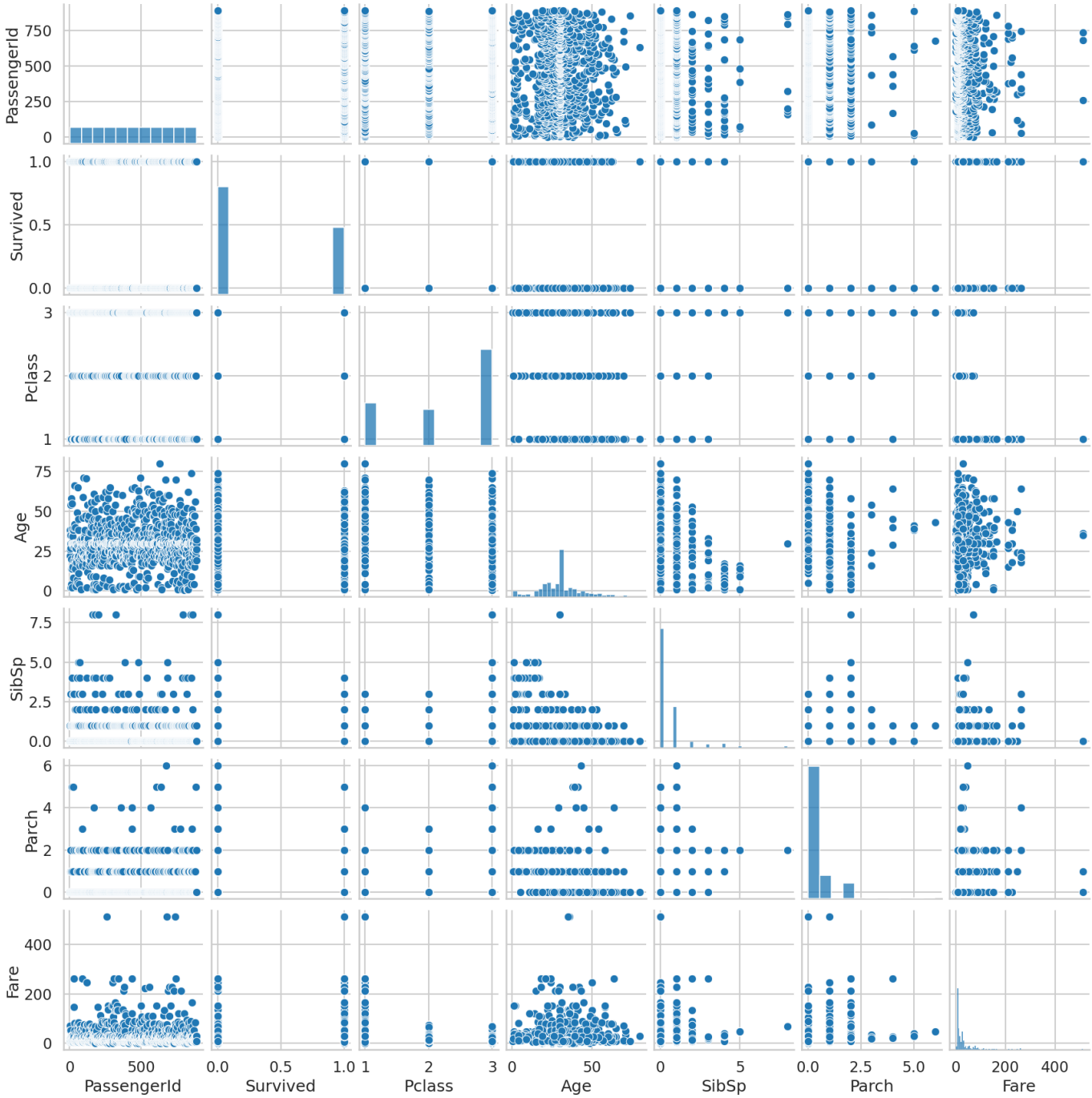
In [230…

```
g = sns.catplot(x= 'Sex', y = 'Age', data=data, kind = 'strip')
plt.show()
```



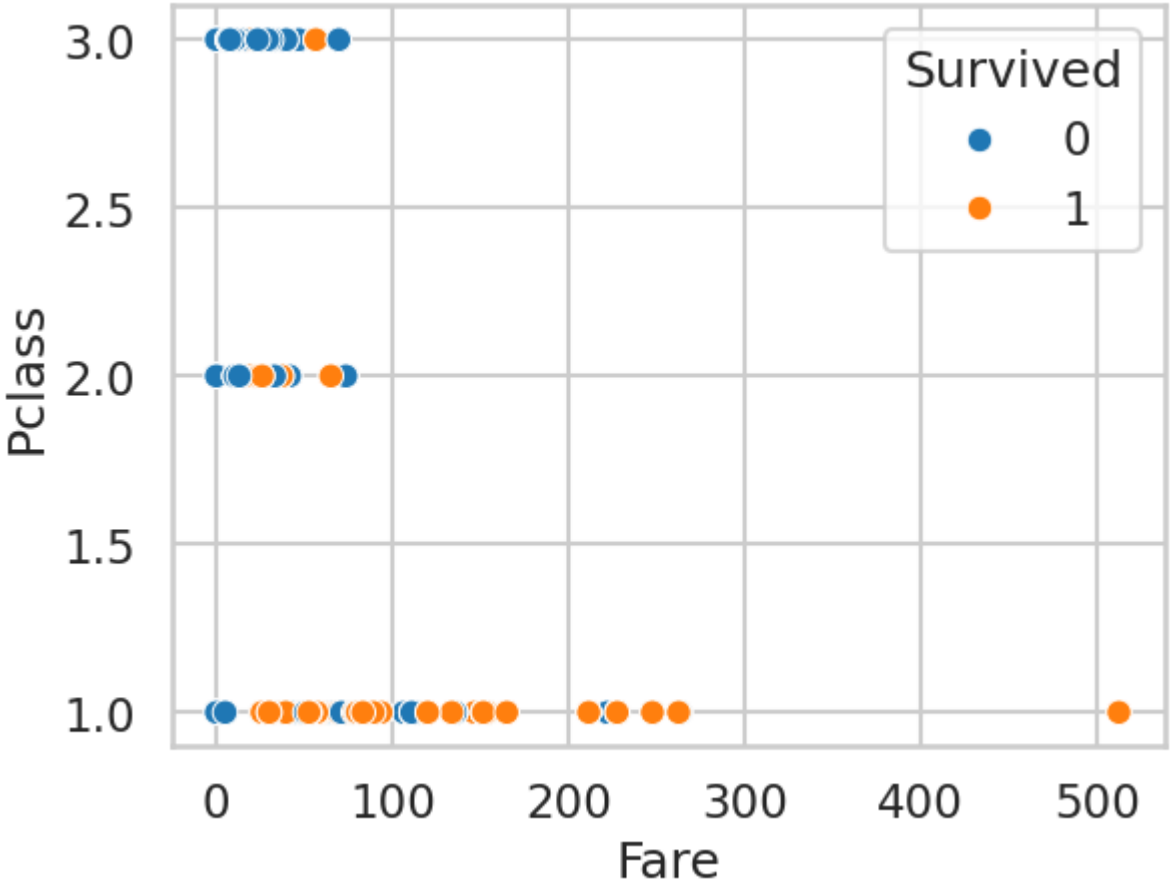
In [231…

```
g = sns.pairplot(data)
plt.show()
```



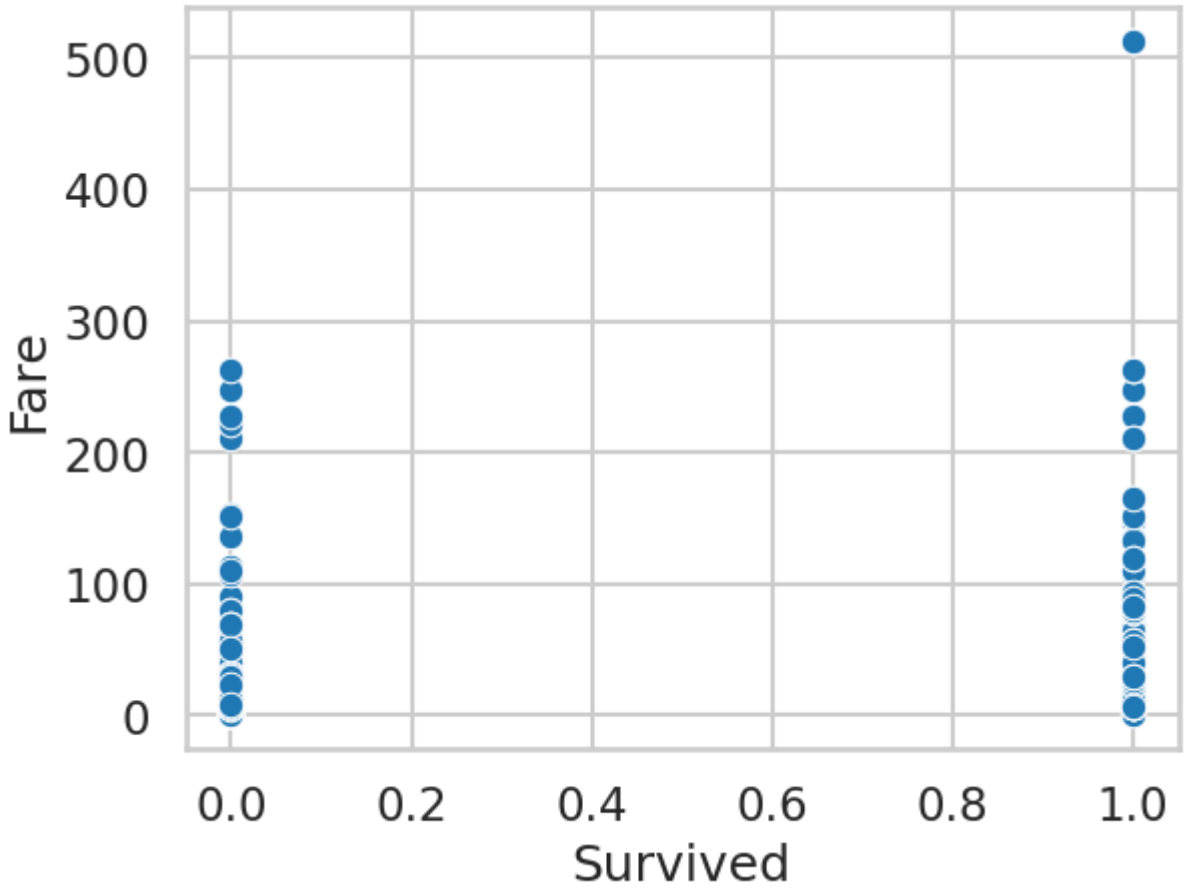
In [232…

```
sns.scatterplot(x = 'Fare', y = 'Pclass', hue = 'Survived', data = data)
plt.show()
```



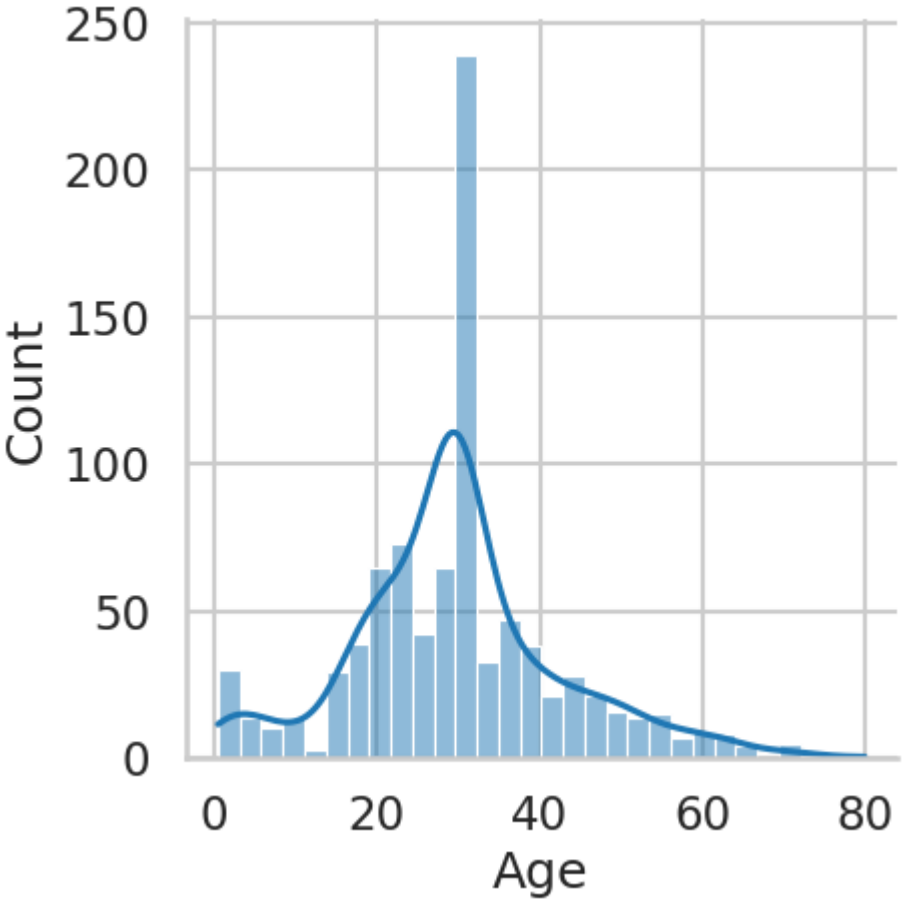
In [233…

```
sns.scatterplot(x = 'Survived', y = 'Fare', data = data)
plt.show()
```



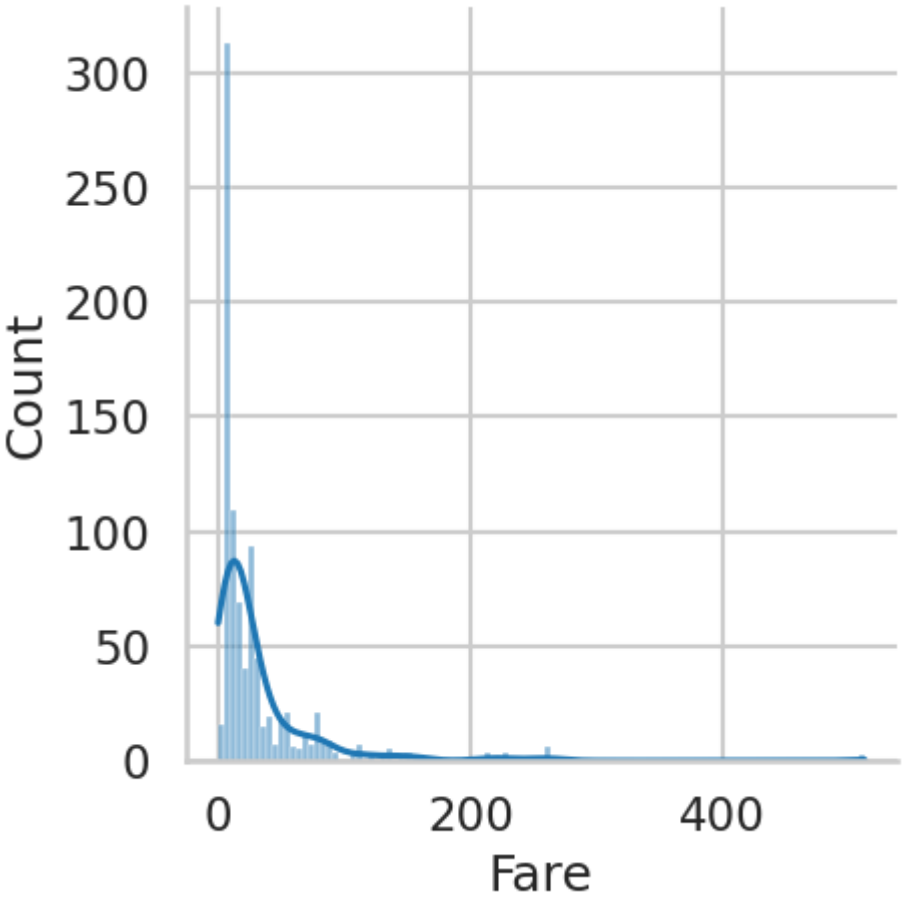
In [234…

```
sns.displot(data['Age'], kde=True)
plt.show()
```



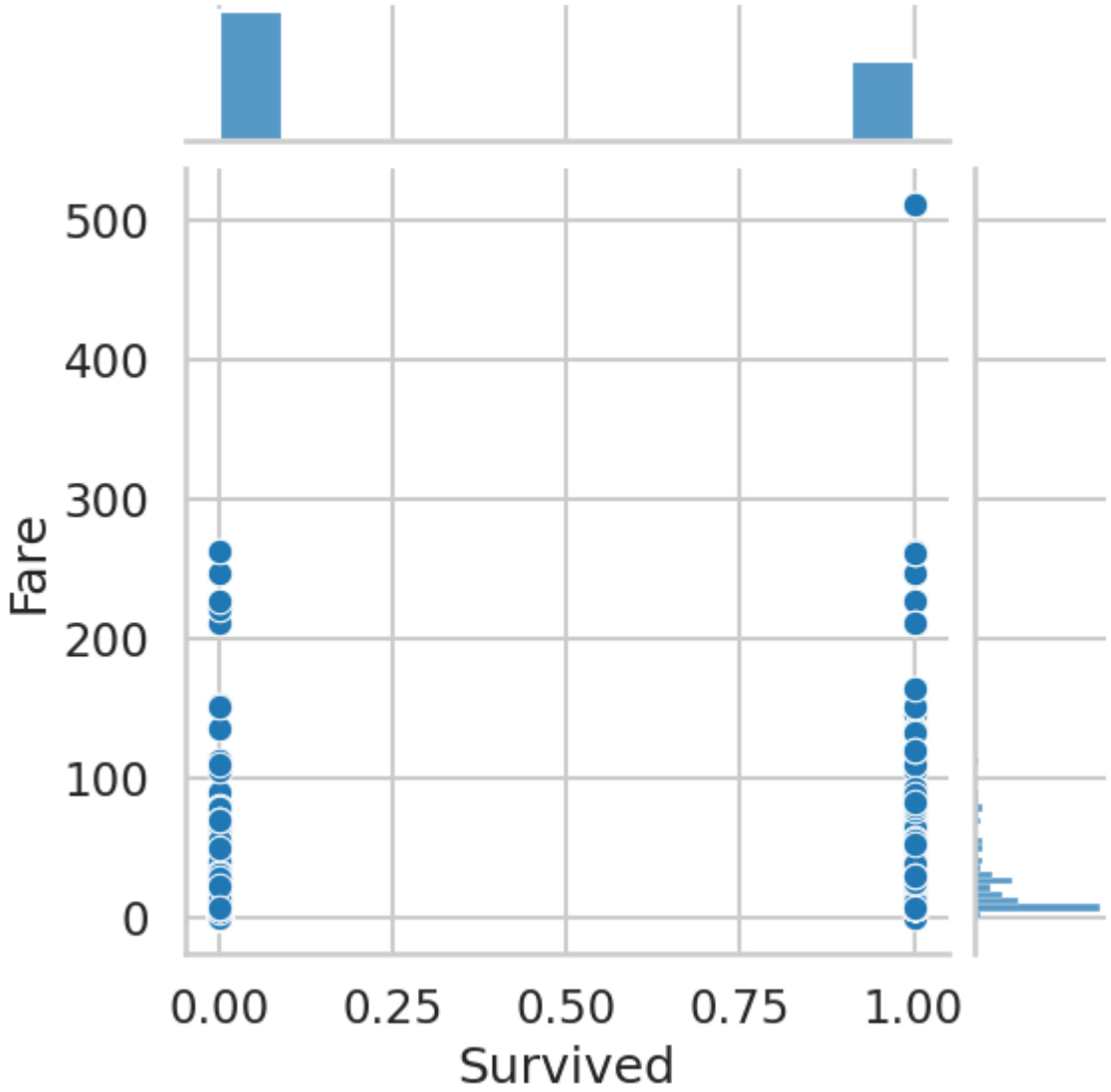
In [235…

```
sns.displot(data['Fare'], kde=True)
plt.show()
```



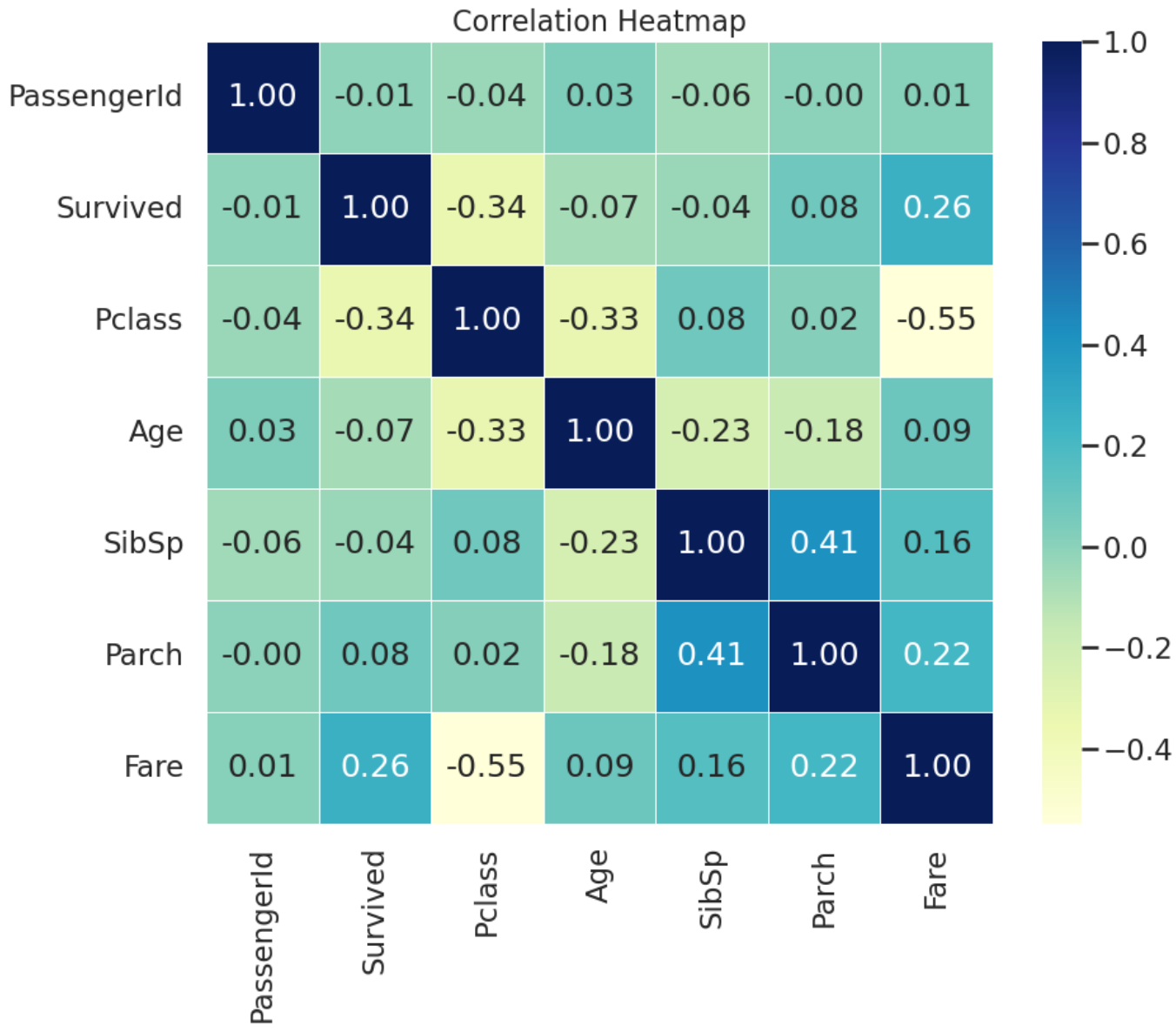
In [236…

```
sns.jointplot(x = "Survived", y = "Fare", kind = "scatter", data = data)
plt.show()
```



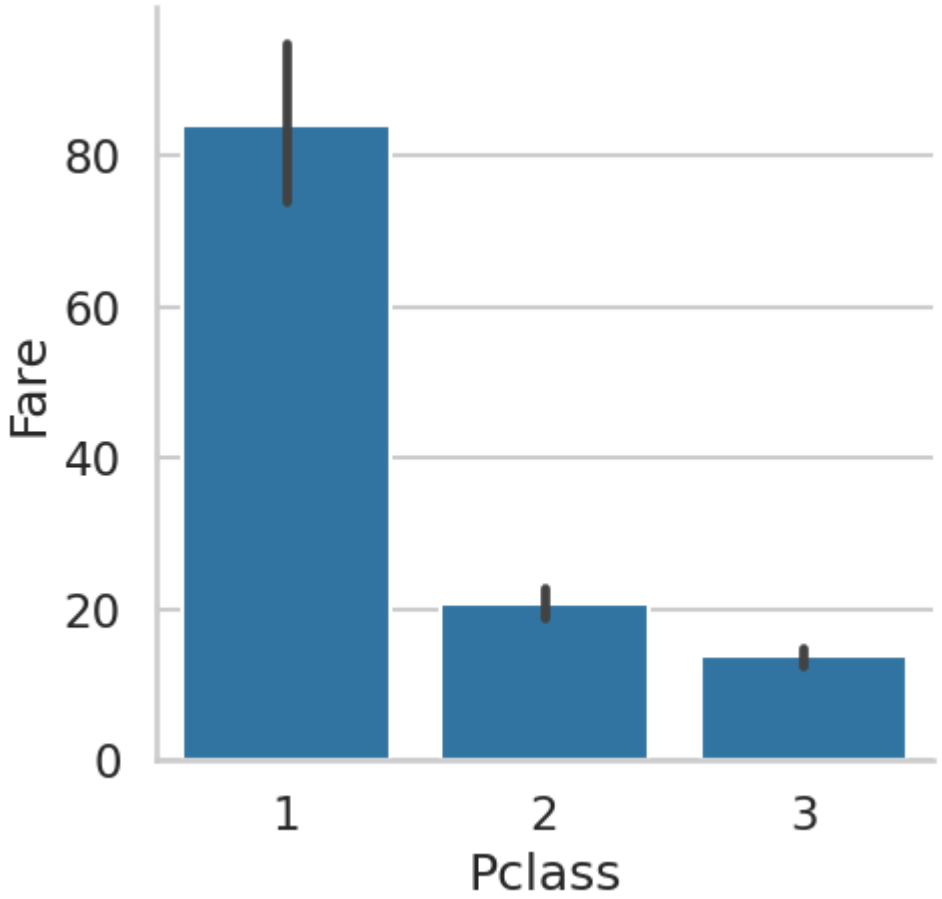
In [237…

```
import seaborn as sns
import matplotlib.pyplot as plt

# Select numeric columns only
numeric_data = data.select_dtypes(include=['number'])

# Compute correlation
tc = numeric_data.corr()

# Plot heatmap
plt.figure(figsize=(10,8))
sns.heatmap(tc, cmap="YlGnBu", annot=True, fmt=".2f", linewidths=0.5)
plt.title('Correlation Heatmap', fontsize=16)
plt.show()
```



In [238…

```
sns.catplot(x='Pclass', y='Fare', data=data, kind='bar')
plt.show()
```



In [ ]: